

CSE 340 Quiz- 5 Section -

Full Marks: 20

Date: Aug 14, 2025

Time: 25 minutes

Name: _____

Roll: _____

1. Convert $1101.1111_2 \times 2^5$ in 32-bit IEEE-754 floating point representation. [5 marks]

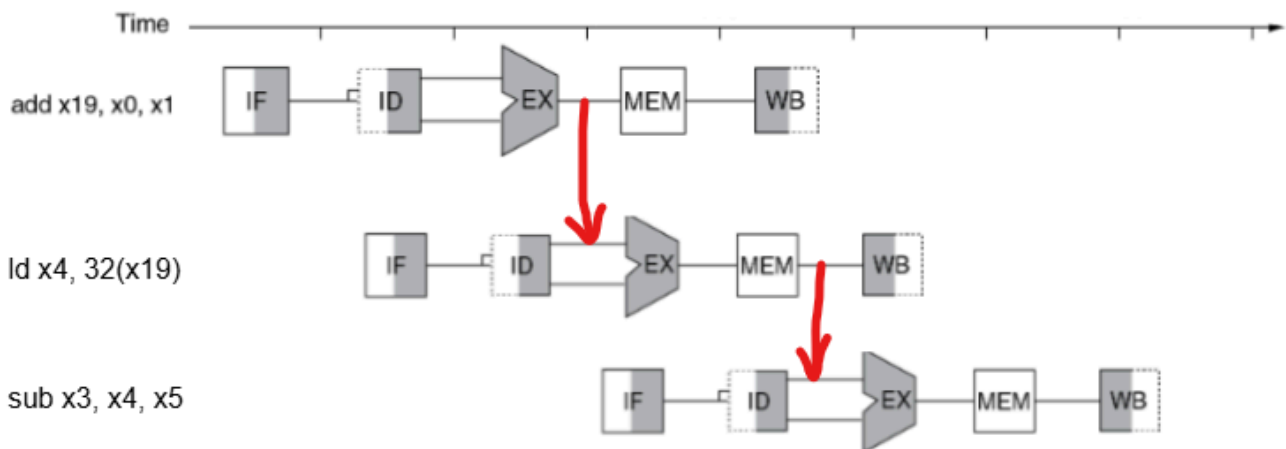
$$1.1011111 \times 2^{5+3} = 1.1011111 \times 2^8$$

- **Sign (S):** 0 (Positive)
- **Exponent (E):** The biased exponent is $8 + 127 = 135$
 - 135 in binary = **10000111**
- **Fraction (F):** The bits after the binary point (padded with zeros to reach 23 bits).
 - **10111110000000000000000**

Final Representation

0	10000111	10111110000000000000000
Sign	Exponent	Fraction

2. Draw the timeline of execution for a pipelined processor to execute the following instructions (the first one is done for you). **Use forwarding where necessary** to speed up execution and calculate the total time.



3. Multiply 9 and 11 using the long multiplication method. Show the steps in the following format. [10 marks]

Iteration	Multiplier	Multiplicand	Product
0	1001	0000 1011	0000 0000
1	1001 1001 0100	0000 1011 0001 0110 0001 0110	0000 1011 0000 1011 0000 1011
2	0100 0100 0010	0001 0110 0010 1100 0010 1100	0000 1011 0000 1011 0000 1011
3	0010 0010 0001	0010 1100 0101 1000 0101 1000	0000 1011 0000 1011 0000 1011
4	0001 0001 0000	0101 1000 1011 0000 1011 0000	0110 0011 0110 0011 0110 0011